

WhatsApp AI Travel Agent

Professional Project Report

A complete end-to-end WhatsApp travel automation built with n8n, WhatsApp Business Cloud, Google Sheets and an AI-powered booking agent.

Project	Premium Travels — WhatsApp AI Travel Agent
Platform	n8n Cloud + WhatsApp Business Cloud
Business Data	Premium_Travels_Client_Google_Sheet_Template
Core Design	Event-driven, stateful, menu-driven automation
AI Layer	Structured Booking AI Agent
Final Status	22 / 22 planned phases completed
Deployment	Published; automatic WhatsApp trigger verified

This report documents the requirements, architecture, workflow, every major feature, data model, AI behavior, testing, production readiness and future expansion.

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1. Executive Summary

Premium Travels — WhatsApp AI Travel Agent is an automated conversational travel-booking system that serves customers through WhatsApp. It moves the customer from the first message through package discovery, package details, AI-assisted information collection, booking confirmation, database storage, office information and human assistance.

The implementation uses n8n Cloud as the orchestration layer, WhatsApp Business Cloud for communication, Google Sheets as the business source of truth, an n8n Data Table for conversational state, and an AI Booking Agent for natural-language collection and validation.

A major design principle is that package data is dynamic rather than hardcoded. Tour, Hajj and Umrah catalogs are maintained in separate sheets, while confirmed bookings are consolidated into one Booking_Requests sheet.

The build followed a Build → Test → Confirm → Next Phase process. Major customer journeys were tested individually and the final published workflow was also tested with an automatic WhatsApp trigger.

2. Project Objectives & Requirements

The system was designed to provide a professional self-service travel experience while preserving deterministic control over business records.

- Automatic response to the customer's first WhatsApp message.
- Welcome + Get Started interaction.
- Direct Main Menu after Get Started.
- Dynamic tour destinations and packages from Google Sheets.
- Separate Hajj and Umrah package catalogs.
- Complete package details and dynamic PDF/media delivery.
- Package actions: Book This Package, Change Package, Main Menu.
- AI collection and validation of six booking details.
- All-six-fields-in-one-message and one-by-one booking collection.
- Maximum ten booking-information attempts.
- Explicit Confirm Booking / Cancel Booking decision.
- Single Booking_Requests sheet for Tour, Hajj and Umrah.
- Dynamic office information from Business_Info.
- Human_Requests record for staff assistance.
- Stateful routing and booking-session memory.
- Automatic production triggering after publication.

3. Technology Stack & Architecture

Component	Purpose
WhatsApp Business Cloud	Inbound events and outbound text, lists, buttons and documents.
n8n Cloud	Workflow orchestration, routing, state, transformations and API calls.
Google Sheets	Dynamic package/business data and operational records.
n8n Data Table — User_State	Per-user state, selected context and booking-session information.
Booking AI Agent	Natural-language booking collection and validation.

Component	Purpose
Structured Output Parser	Predictable JSON contract for the AI.
Simple Memory	Conversation context keyed by booking_session_id.

High-Level Architecture

```

Customer WhatsApp
  ↓
WhatsApp Trigger
  ↓
Get User State
  ↓
Prepare User Context
  ↓
Refresh User Activity
  ↓
Restore User Context
  ↓
Message Router
  ├── Get Started / Main Menu
  ├── Tour Packages
  ├── Hajj Packages
  ├── Umrah Packages
  ├── Office Location
  ├── Talk to Human
  └── Booking AI Agent

```

4. Complete Customer Journey

FIRST MESSAGE
 Hi / hello / Assalam o Alaikum / any first message
 ↓
 Welcome + Get Started
 ↓
MAIN MENU
 ├── Explore Tour Packages
 ├── Hajj Packages
 ├── Umrah Packages
 ├── Office Location
 └── Talk to a Human

TOUR
 Main Menu → Dynamic Destinations → Destination → Dynamic Packages
 → Package Details + Media → Book / Change Package / Main Menu

HAJJ
 Main Menu → Hajj Package List → Package Details + Media
 → Book / Change Package / Main Menu

UMRAH
 Main Menu → Umrah Package List → Package Details + Media
 → Book / Change Package / Main Menu

BOOKING
 Book Package → AI Collection → Summary → Confirm / Cancel
 → Booking_Requests if confirmed

OTHER SERVICES
 Office → Business_Info → Details → Main Menu
 Human → Human_Requests → Handoff → Main Menu

The customer experience is menu-driven at navigation points and AI-assisted where natural-language interaction is useful: booking information collection and validation.

5. n8n Workflow & Message Routing

Rule / State	Action
main_menu	Send Main Menu
cancel_booking	Cancel Booking State
confirm_booking	Booking save route

Rule / State	Action
COLLECTING_BOOKING_INFO	Booking AI Agent
explore_tours	Get Travel Packages
hajj_packages	Get Hajj Packages
umrah_packages	Get Umrah Packages
destination_*	Save Selected Destination
package_*	Get Selected Package
hajj_package_*	Get Selected Hajj Package
umrah_package_*	Get Selected Umrah Package
office_location	Get Office Info
talk_to_human	Save Human Request

Routing expressions use optional chaining for interactive IDs so plain text messages can be handled safely.

```
{{ $json.messages[0].interactive?.list_reply?.id }}
```

Specific booking rules are evaluated before generic text handling so the AI booking conversation is not accidentally routed back to the Main Menu.

6. User State & Session Management

Field	Purpose
phone_number	WhatsApp user identifier used for matching.
state	Current conversational state.
selected_destination	Selected tour destination or Hajj/Umrah category.
selected_package_id	Current selected package.
booking_attempts	Number of booking-information messages/attempts.
booking_session_id	Unique AI conversation session key.
updatedAt	Activity timestamp.

Booking sessions remain active for up to 24 hours. Normal browsing is treated as a new session after 15 minutes of inactivity. The booking session ID is generated from the WhatsApp number and timestamp.

```
{{ $('WhatsApp Trigger').item.json.contacts[0].wa_id + '-booking-' + Date.now() }}
```

7. Tour Package Module

Travel_Packages is the source of truth for tour destinations and packages. The workflow reads rows, extracts unique destinations, builds a dynamic WhatsApp list, saves the selected destination and retrieves only packages for that destination.

Dynamic destination extraction:

```
const destinations = [...new Set(
  $input.all().map(item => item.json.Destination).filter(Boolean)
)];

return destinations.map((destination, index) => ({
  json: { destination_number: index + 1, destination }
}));
```

Dynamic package list:

```
const packages = $input.all();

return [{
  json: {
    packages: packages.map((item, index) => ({
```

```

id: `package_${index + 1}`,
title: item.json.Package_Name,
description: `${item.json.Duration} • ${item.json.Price} ${item.json.Currency}`,
package_number: index + 1
}))
}
}];

```

Package details are built entirely from the selected Google Sheets row. The document/media is sent first, a five-second wait preserves message order, and then details and the three package actions are sent.

8. Hajj Package Module

Hajj is maintained in the separate Hajj_Packages sheet. The Main Menu ID hajj_packages opens the dynamic list; individual package IDs use hajj_package_*. The selected package is retrieved, saved in User_State with selected_destination = Hajj, and formatted into a package detail response.

The Hajj branch sends document media, waits five seconds, sends details and displays Book This Package, Change Package and Main Menu. Hajj booking reads its package source from Hajj_Packages and normalizes the result for the common Booking_Requests save path.

9. Umrah Package Module

Umrah uses the separate Umrah_Packages sheet and mirrors the Hajj architecture. The Main Menu ID umrah_packages opens the list and individual selections use umrah_package_*.

The selected Umrah package is loaded from Umrah_Packages, saved into User_State with selected_destination = Umrah, converted into normalized booking data and passed to Booking_Requests.

During testing, the Umrah booking lookup was initially pointed at Booking_Requests. This was corrected to Umrah_Packages and the test passed. The issue was caught before final production verification.

10. AI Booking Agent

The Booking AI Agent collects and validates booking information but does not directly write to Google Sheets. Deterministic workflow nodes control confirmation, routing and the actual save.

Required information: Full Name, CNIC Number, Passport Validity, Departure City, Travel Month and Number of Travelers.

The agent supports all six details in one message or a fixed one-by-one sequence. It produces structured output and, when valid, a final summary followed by two buttons: Confirm Booking and Cancel Booking.

Structured output contract:

```

{
  "response": "string",
  "full_name": "string|null",
  "cnic": "string|null",
  "passport_validity": "string|null",
  "departure_city": "string|null",
  "travel_month": "string|null",
  "number_of_travelers": "integer|null",
  "booking_ready": "boolean",
  "confirmed": "boolean"
}

```

The AI is constrained not to invent package data, Booking IDs or claim that a booking has been saved before the workflow actually performs the save.

11. Booking Validation & 10-Attempt Protection

The workflow controls the attempt counter. Every incoming booking-information message counts as one attempt, including invalid information. Supplying all six fields in one message counts as one attempt.

```

booking_attempts >= 10
AND booking_ready == false
↓
Return customer to Main Menu

Otherwise
↓
Continue booking conversation

```

If valid information is received on attempt ten, the booking continues. The protection prevents an incomplete conversation from remaining active indefinitely.

12. Booking Confirmation & Saving

A customer must explicitly select Confirm Booking before a booking is written as Confirmed. Cancel Booking exits the process without creating a confirmed booking.

Branch	Condition	Package Source
Tour	selected_package_id does not start with HU-	Travel_Packages
Hajj	starts with HU-HAJ-	Hajj_Packages
Umrah	starts with HU-UMR-	Umrah_Packages

All three branches converge on the same Booking_Requests sheet. Stored information includes Booking_ID, Created_At, Customer_Name, WhatsApp_Number, Package_ID, Destination, Package_Name, Duration, Price, Departure_City, Travel_Month, Travelers, CNIC, Passport_VValidity, Status and Notes where applicable.

```
{{ 'PT-' + Date.now() }}
```

13. Office Location

Office Location reads the Business_Info sheet dynamically, keeping address, contact information, office hours and map information outside the workflow code.

```

Main Menu
↓ office_location
Get Office Info
↓
Build Office Location Message
↓
Send Office Location
↓
Send Office Location Actions
↓
Main Menu

```

The response can include Business Name, Office Address, Phone, WhatsApp, Email, Office Hours and Google Maps URL.

14. Talk to a Human

Talk to a Human creates an operational record in Human_Requests and then confirms the handoff to the customer.

Field	Configured Behavior
Request_ID	HR- + timestamp
Created_At	Current workflow timestamp
Customer_Name	Blank because this menu action does not collect a name
WhatsApp_Number	Captured from WhatsApp Trigger wa_id
Reason	Customer requested human assistance
State	HUMAN_REQUEST
Status	Pending

Field	Configured Behavior
Staff_Notes	Blank for staff follow-up

Main Menu → Talk to a Human → Save Human Request
→ Send Human Handoff Message → Send Human Actions → Main Menu

15. Google Sheets Data Architecture

Sheet	Purpose
Travel_Packages	Tour destinations and tour package catalog.
Hajj_Packages	Hajj package catalog.
Umrah_Packages	Umrah package catalog.
Booking_Requests	Single confirmed-booking register for all package categories.
Human_Requests	Customer requests for staff assistance.
Business_Info	Office/business contact and location information.
README	Workbook guidance.

This structure separates maintainable business data from reusable workflow logic. Staff can update package information in Sheets without rebuilding individual package nodes.

16. WhatsApp API & Media

WhatsApp Business Cloud is used for customer-facing communication. The workflow sends text, interactive buttons, interactive lists and document media through the Graph API.

Interactive features include Get Started, Main Menu, dynamic destination lists, dynamic package lists, package actions, confirmation/cancellation controls and Main Menu return buttons.

Package media is sent as a WhatsApp document using Media_URL from the relevant package sheet. A five-second wait is used before package details. Production media links must be accessible to WhatsApp; placeholder/example links must be replaced before real customer use.

17. Testing & Verification

Area	Result
WhatsApp Trigger	Passed — automatic event trigger verified.
Get Started	Passed.
Main Menu	Passed.
Tour destination/package flow	Passed.
Tour booking save	Passed.
Hajj list/details/media	Passed.
Hajj booking save	Passed.
Umrah list/details/media	Passed.
Umrah booking save	Passed after source-sheet correction.
AI validation	Passed.
Ten-attempt protection	Passed.
Confirm / Cancel	Passed.
Office Location	Passed.
Talk to Human	Passed.

Area	Result
Operational sheet cleanup	Passed.
Automatic production trigger	Passed.
Final end-to-end test	Passed.

The final production-style test confirmed that the workflow responds automatically without manual Execute Workflow actions.

18. Security & Production Readiness

- Working WhatsApp and Google Sheets credentials were preserved rather than unnecessarily replaced.
- API credentials are referenced through n8n credentials rather than customer-facing message content.
- Booking confirmation is required before writing a Confirmed booking.
- Cancellation exits the booking path without saving a booking.
- AI output is constrained through a structured parser.
- AI is prevented from claiming a booking is saved before the deterministic save completes.
- Booking sessions have defined activity windows.
- Ten-attempt protection prevents stalled booking sessions.
- Operational test rows were cleaned from Human_Requests and Booking_Requests was checked.
- The workflow was published and automatic WhatsApp triggering was verified.

19. 22-Phase Completion

Phase	Module	Status
1	WhatsApp Trigger	Complete
2	Get Started	Complete
3	Main Menu	Complete
4	User State / Session	Complete
5	Message Router	Complete
6	Dynamic Tour Destinations	Complete
7	Tour Package List	Complete
8	Tour Details + Media	Complete
9	Tour Actions / Navigation	Complete
10	Booking AI Agent	Complete
11	Booking Information Collection	Complete
12	Validation + 10 Attempts	Complete
13	Confirmation / Cancellation	Complete
14	Tour Booking Save	Complete
15	Hajj Packages + Booking	Complete
16	Umrah Packages + Booking	Complete
17	Office Location	Complete
18	Talk to Human	Complete
19	Human Request Save + Handoff	Complete

Phase	Module	Status
20	Full End-to-End Testing	Complete
21	Production / Auto Trigger Publish	Complete
22	Final Cleanup + Production Readiness	Complete

Final status: 22 / 22 planned phases completed. The workflow is published and automatic triggering has been verified.

20. Future Enhancements

- Staff notification integration for Human_Requests.
- Automated booking-status updates and customer follow-up.
- Admin dashboard for package availability and booking statistics.
- Payment/deposit integration with appropriate security and reconciliation controls.
- Multilingual AI responses.
- Automated reminders for passport validity and travel dates.
- Centralized logging and alerting.
- Analytics for package interest, bookings and human handoffs.

These are optional enhancements and are not prerequisites for the completed core project.

21. Conclusion

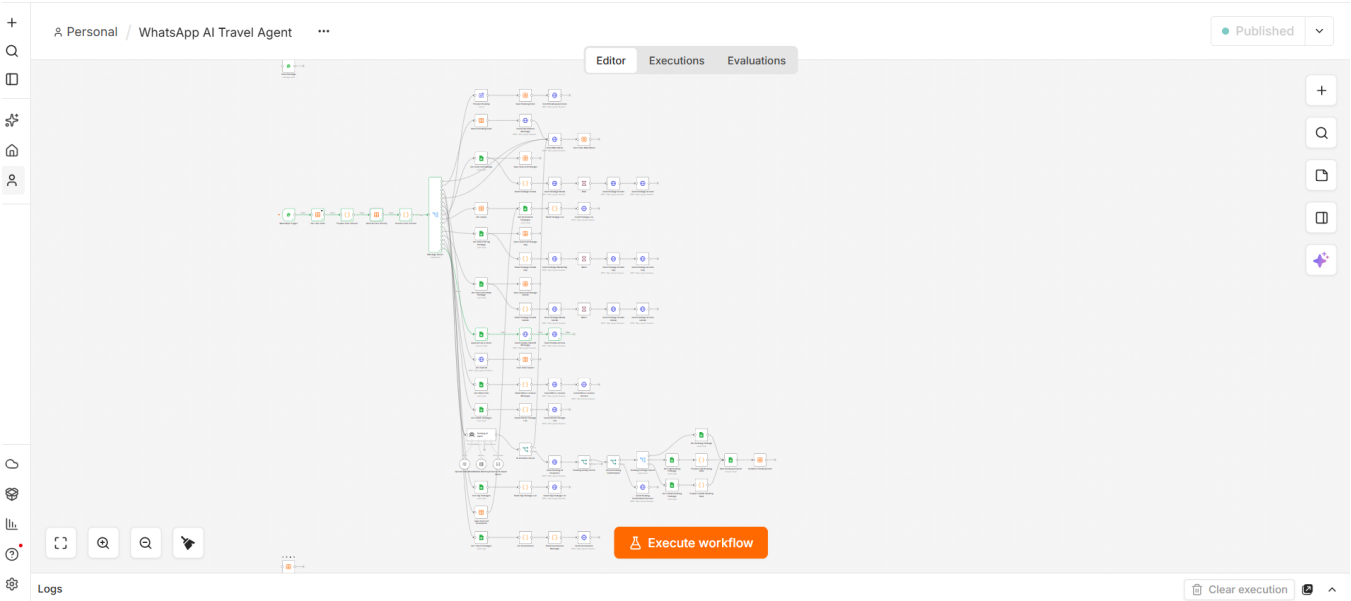
Premium Travels now has a complete WhatsApp-based AI travel automation combining deterministic workflow automation with AI-assisted customer interaction. The system separates business data from workflow logic, supports dynamic Tour, Hajj and Umrah catalogs, captures confirmed bookings in a unified operational sheet, and provides both self-service and human-assistance paths.

The project demonstrates practical use of n8n orchestration, WhatsApp Business Cloud APIs, Google Sheets as a lightweight operational database, stateful conversational routing, structured AI outputs and production-oriented testing. It is therefore more than a simple chatbot: it is a connected travel-booking workflow with defined states, business records, confirmation controls and automated customer journeys.

With all 22 planned phases completed and the published automatic trigger verified, the system is ready to serve as a professional portfolio, academic/project submission or foundation for further business integrations.

Appendix A — Complete n8n Workflow Visual

Actual workflow screenshot from the completed Premium Travels WhatsApp AI Travel Agent implementation.



Visual reference: The screenshot shows the central WhatsApp Trigger → User State → Message Router architecture and the connected Tour, Hajj, Umrah, Booking, Office Location and Human Assistance branches.